

Week 6: Feature Types

AI Frameworks

Today's learning goals

By the end of this lecture you should be able to:

1. Distinguish **numerical**, **categorical (nominal)**, and **ordinal** features
2. Explain why type matters for plots and models
3. Describe simple encoding (codes / one-hot) at an intuition level
4. Explain why feature **scale** matters for distance-based ideas like k-NN

Agenda (80 minutes)

1. Features vs target (recap)
2. Numerical features
3. Categorical vs ordinal
4. Why types matter
5. Encoding ideas
6. Scaling intuition (k-NN preview)

Words we will use

- **Features / inputs:** columns we use to predict
- **Target / label:** the column we want to predict

Today focuses on *types of features*—not training a model yet (Week 7+).

Numerical features

Numbers with mathematical meaning:

- **Continuous:** height, temperature, price (can take many real values)
- **Discrete:** counts — number of rooms, visits (whole numbers)

You can sensibly compute means, differences, and plot histograms/scatters.

Careful: not every number is “numeric meaning”

- Zip codes, jersey numbers, encoded IDs — digits, but not quantities
- Treat those as categorical (or IDs), not as sizes to average

Common mistake

Averaging postcodes or taking “mean student id”.

Categorical (nominal)

Labels without a natural order:

- Colour: red / blue / green
- City name; blood type; favourite subject

“Red” is not less than “blue”.

Categories **with** a meaningful order:

- Size: $S < M < L$
- Rating: poor < ok < good < excellent
- Education level (sometimes)

Order matters; exact gaps between levels may still be unclear.

Quick sorting hat

Example	Type
Age in years	Numerical
Temperature °C	Numerical
Eye colour	Categorical (nominal)
T-shirt size S/M/L	Ordinal
Product ID 481516	ID / categorical (not a quantity)

Quiz 1 — Types

Quiz

Survey answer “disagree / neutral / agree” is best seen as:

1. Continuous numerical in kilometres
2. Ordinal categorical
3. A matplotlib colour map
4. Unsupervised clustering only

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

2. Ordinal — ordered categories; not a plain physical measurement.

For plots

- Numeric → histogram, scatter, line
- Categorical → bar counts, box plots by group
- Wrong type → nonsense charts (“mean of city name”)

For models (preview)

- Many models expect numbers
- Categories usually need encoding first
- Distance-based methods care about feature scale

Why encode?

Algorithms often need numbers. Categories must be turned into numeric form **without inventing fake order** (for nominal data).

Two simple ideas

- **Category codes:** map each label to an integer (0,1,2,...)
Fine for trees sometimes; risky for distance models if nominal
- **One-hot:** new 0/1 column per category
“is_red”, “is_blue”, ... — no fake ranking

```
1 # category codes (integers)
2 df["city_code"] = df["city"].astype("category").cat.codes
3
4 # one-hot
5 dummies = pd.get_dummies(df["city"], prefix="city")
6 df2 = pd.concat([df, dummies], axis=1)
```

What & why — not theory

- Goal: make categories usable by numeric algorithms
- One-hot avoids implying $S < M < L$ when colours have no order
- For true ordinal data, ordered codes can sometimes be OK—be intentional

Quiz 2 — Encoding

Quiz

For nominal colours red/blue/green, why is one-hot often preferred over codes 0,1,2 for a distance-based model?

1. Because blue is always 2
2. Codes invent a fake order/distance between colours
3. One-hot deletes the dataset
4. Colours cannot be columns

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

2. Integer codes imply false numeric relationships; one-hot treats categories as separate flags.

Scale problems (picture in words)

Suppose features are:

- Age: about 10–18
- Annual income in local currency: 0–100000

Distance “as the crow flies” in feature space will be dominated by income unless we rescale—important for k-NN (coming soon).

Scaling as a black box

Common simple ideas:

- **Min-max:** squeeze a column into roughly 0–1
- **Standardise:** centre near 0 with spread near 1 (StandardScaler)

You do not need the formulas memorised—know *why* we scale.

sklearn preview (optional awareness)

```
1 from sklearn.preprocessing import StandardScaler
2 scaler = StandardScaler()
3 X_scaled = scaler.fit_transform(X)  # X = numeric feature matrix
```

Practice may try this as a black box after encoding.

Quiz

True or false?

If one feature is in huge units and another is tiny, a distance-based method may mostly “see” the huge feature unless we rescale.

1. True
2. False

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

True. That is the core scaling intuition for k-NN-style distance.

Key takeaways

1. Numerical vs categorical (nominal) vs ordinal
2. Types guide plots and preparation for models
3. Encoding turns categories into numbers carefully
4. Scaling balances feature influence for distance methods

Next up

Practice: label column types; encode 1–2 categoricals; optional scaler.

Week 7: train/test split and baselines.